

HGLM FOR ORDINAL AND NOMINAL OUTCOMES



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Ordinal Variables

An ordinal variable is a categorical variable where the categories are ordered in some logical way.

They generally measure some underlying latent variable that cannot be (or is not easily) measured directly.

We assume this *latent variable* is continuous. It being higher or lower will change the probability that a unit will fall into a particular category.

Ordinal Variables

For example, assume we ask students how often they do their homework, and give them the option of responding: (1) Rarely, (2) Usually, or (3) Always.

Students that are more diligent are more likely to choose higher (versus lower) categories. Interventions that improve diligence will result greater chances of students choosing higher (versus lower) categories.

Ordinal Variables

In the textbook example, teachers were asked if they would choose teaching again if they could start all over. They could answer “Yes,” “Not sure,” or “No.”

Ordered Logistic Regression

We therefore estimate the *log odds of belonging to a specific category or lower* (versus a higher category).

$$E(Y) = \ln\left(\frac{\Phi_{R \leq m}}{1 - \Phi_{R \leq m}}\right) = \ln\left(\frac{\Phi_{R \leq m}}{\Phi_{R > m}}\right)$$

Where m is an ordered category numbered 1, 2,..., M

Ordinal Logistic Regression

Note that this model can be estimated in the opposite way. You could instead, estimate the log odds of being in each category *or higher*.

The HLM software estimates log odds of being in a category or lower though, which is somewhat more common.

Interpretation

You will estimate $M-1$ intercepts in your model. Each intercept (often called a cut point) represents the log odds of the outcome being in the m^{th} category or lower.

In our example, there will be two intercepts, one representing the cut point between (1) Rarely and (2) Usually, and between (2) Usually and (3) Always.

So therefore we'll estimate...

$$\ln \left(\frac{\phi_{R \leq m}}{1 - \phi_{R \leq m}} \right) = \beta_{0j} + \beta_{1j}X_{1ij} + \beta_{2j}X_{2ij} + \cdots + \beta_{kj}X_{kij} \\ + \delta_m \text{ if } m > 1$$

And we will have two intercepts (called “thresholds” in the HLM text, also called “cut points”),

$$\theta_1 = \beta_{0j} \text{ and}$$

$$\theta_2 = \beta_{0j} + \delta_2$$

Interpretation

θ_1 represents the average log odds that a student will be in the lowest category (Rarely) when all covariates are 0.

θ_2 represents the average log odds that a student will be in the second category *or lower* (Usually or Rarely) when all covariates are 0.

Interpretation

Coefficients represent the change in the log odds of being in a higher category (versus a lower one) for a one-unit change in the predictor.

You can also calculate odds ratios, which represent the multiplicative change in odds of moving into a higher category for a one-unit change in the predictor.

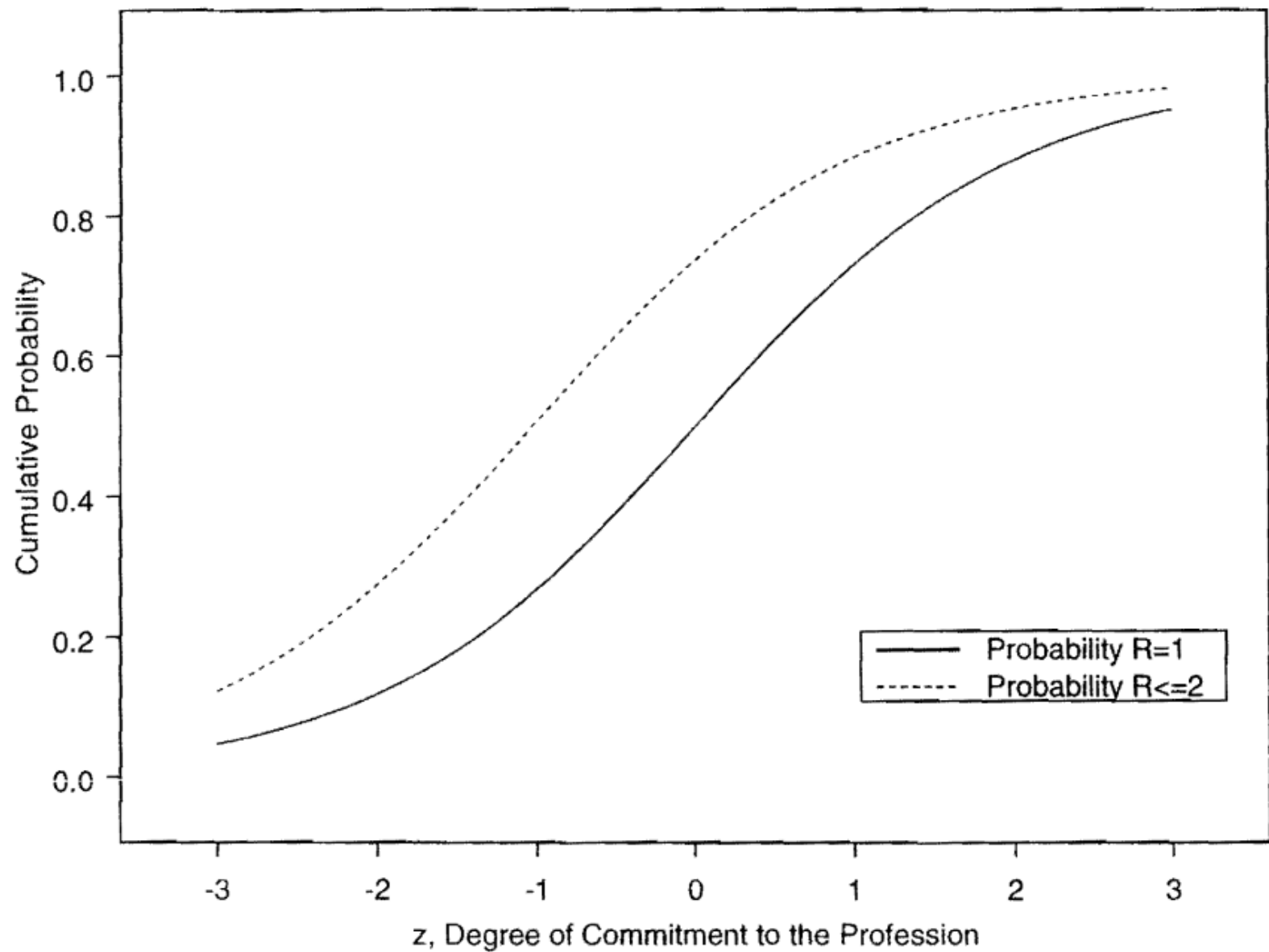


Figure 10.5. Association Between Latent Degree of Commitment (horizontal axis) and Cumulative Probability (vertical axis)

Ordered Logistic Regression

You can calculate the probability of a particular response (or lower) with

$$\Phi_{R \leq m} = \frac{e^{\beta_{0j} + \beta_{1j}X_{1ij} + \beta_{2j}X_{2ij} + \dots + \beta_{kj}X_{kij}}}{1 + e^{\beta_{0j} + \beta_{1j}X_{1ij} + \beta_{2j}X_{2ij} + \dots + \beta_{kj}X_{kij}}}$$

Notice that positive coefficients indicate increases in the odds of higher values (and lower odds of lower values).

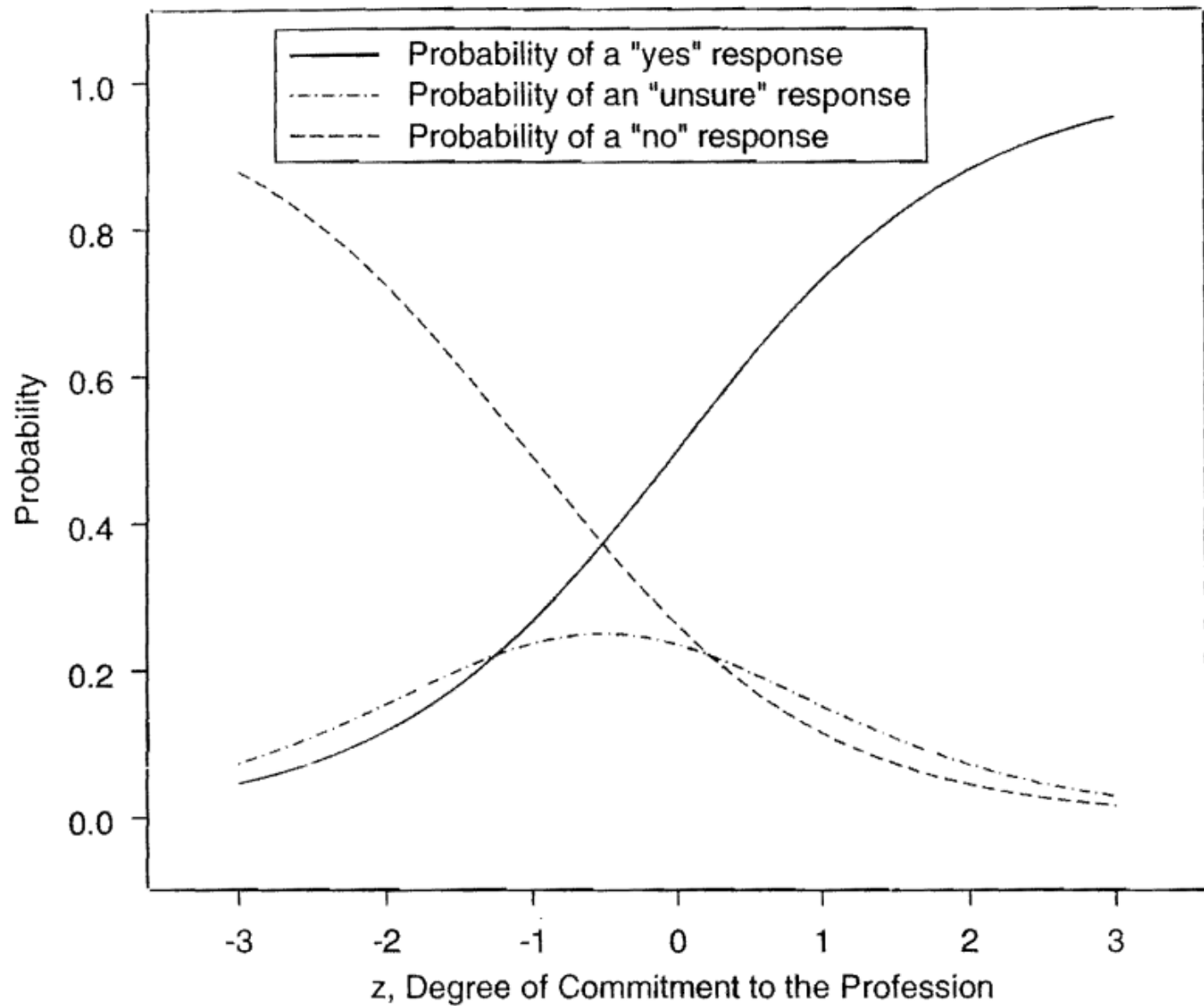


Figure 10.4. Association Between Latent Degree of Commitment (horizontal axis) and Probabilities of Responding “Yes,” “Unsure”, or “No” (vertical axis)

Visualization

Using smart visualization is especially helpful in communicating effect estimates of these complex models.

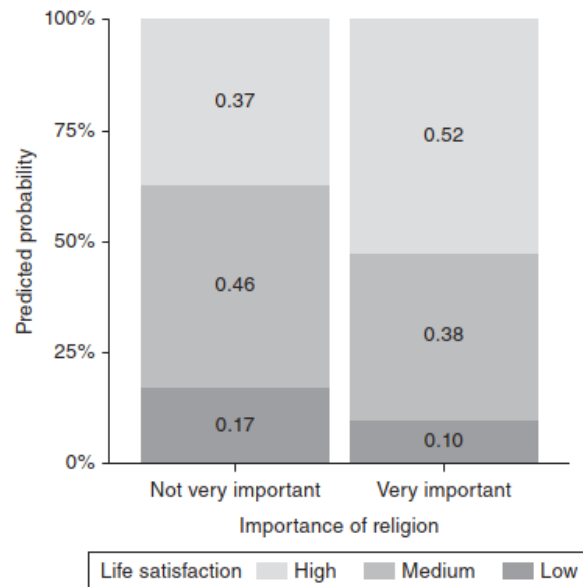


Figure 3.1 Predicted probabilities of low, medium and high life satisfaction from a proportional odds ordinal regression on subjective importance of religion (from Model 3.1)

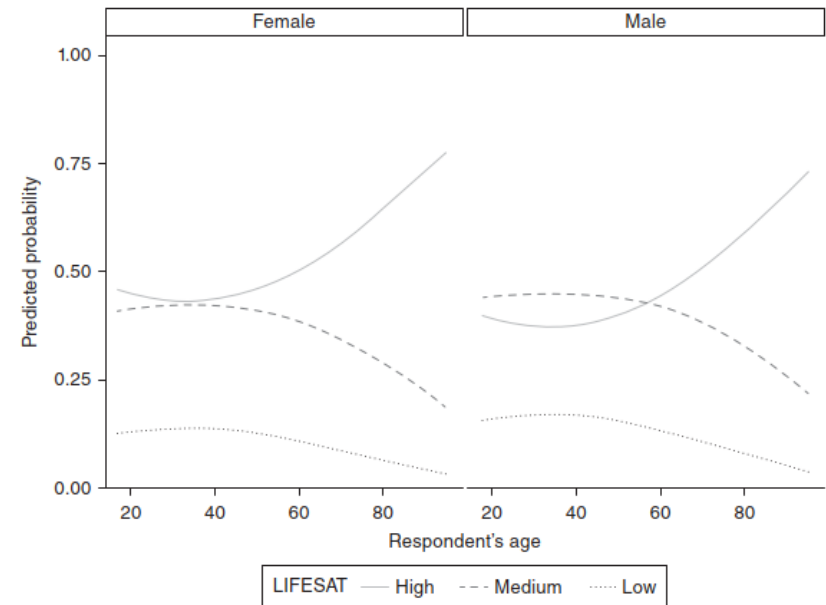


Figure 3.3 The relationship between life satisfaction, age and gender, predicted by Model 3.4

Tip: Use **shades of a single color** to distinguish **quantitatively-ordered categories**, such as ordinal outcomes.

Proportional Odds Assumption

The most popular type of ordered logistic regression assumes that changes in a predictor will have a similar effect on the log-odds of moving up into each category.

For example, the effect an intervention on homework completion will be similar for moving kids from “Rarely” to “Usually” as it is for “Usually” to “Always.”

Proportional Odds Assumption

You can examine the proportional odds assumption by:

- Dichotomizing your outcome around each cutpoint, then
- Running logistic regressions with your model predicting each dichotomous outcome, then
- Comparing the coefficients to see if they are similar

Proportional Odds Assumption

If your model fails the proportional odds assumption, you can

- Run a multinomial logistic regression (described in slides below)
- Treat your outcome as binary or continuous

Do I *Have* To?

Running and interpreting an ordered logistic regression model is considerably more complex than a normal HLM model.

It does not make a linearity assumption between predictors and the outcome, but it can come with other assumptions (e.g. proportional odds).

Do I *Have To*?

This is why many analysts prefer to treat ordinal outcomes as continuous if we possibly can.

My recommendation: Run both HLM and the ordered logit models. See if the results are substantively similar.

- Similar effect sizes relative to each other
- Similar statistical significance of coefficients

If so, go with the simpler model.

Multinomial Logistic Regression

Nominal variables are categorical, and do not have ordered categories (e.g. race, political affiliation, school subjects).

We can use a logistic model to estimate the log odds of one outcome relative to reference category.

Link function is:

$$E(Y = m) = \ln \left(\frac{\phi_m}{\phi_M} \right) = \ln \left(\frac{\text{Prob}(R = m)}{\text{Prob}(R = M)} \right)$$

Multinomial Logistic Regression

Consider an outcome with three categories. We pick a base category, say $Y=3$.

We estimate:

$$\ln \left(\frac{\phi_{R=1}}{\phi_{R=3}} \right) = \beta_{0j(1)} + \beta_{1j(1)}X_{1ij} + \cdots + \beta_{kj(1)}X_{kij}$$

$$\ln \left(\frac{\phi_{R=2}}{\phi_{R=3}} \right) = \beta_{0j(2)} + \beta_{1j(2)}X_{1ij} + \cdots + \beta_{kj(2)}X_{kij}$$

Reference category

You should be deliberate in the choice of a reference category. Choose one that represents a theoretical default category to ease interpretation.

Underlying probabilities will be the same regardless of which is chosen, but the coefficients and risk ratios will be different.

In particular, avoid choosing a category that is very scarce.

Multinomial Logistic Regression

You can back out the probabilities from this type of model as well, but it is complicated.

$$\phi_{R=1} = \frac{e^{\beta_{0j(1)} + \dots + \beta_{kj(1)} X_{kij}}}{1 + e^{\beta_{0j(1)} + \dots + \beta_{kj(1)} X_{kij}} + e^{\beta_{20} + \beta_{21} X_{1i} + \dots + \beta_{2k} X_{ki}}}$$

$$\phi_{R=2} = \frac{e^{\beta_{0j(2)} + \dots + \beta_{kj(2)} X_{kij}}}{1 + e^{\beta_{0j(1)} + \dots + \beta_{kj(1)} X_{kij}} + e^{\beta_{0j(2)} + \dots + \beta_{kj(2)} X_{kij}}}$$

$$\phi_{R=3} = \frac{1}{1 + e^{\beta_{0j(1)} + \dots + \beta_{kj(1)} X_{kij}} + e^{\beta_{0j(2)} + \dots + \beta_{kj(2)} X_{kij}}}$$

Relative Risk Ratios

You can get something like an odds ratio for multinomial logit models. They describe the relative risks of being in one category versus the reference category, conditional on a change in a predictor.

These are often part of the default output (or a standard option) in most stat packages capable of running this model. R does not do this, but I have provided code for doing so.

Risk Ratios

The odds (ratio of probabilities) of being in a specific category versus the reference category are called **risk ratios (RR)**.

$$RR_{1/3} = \frac{\phi_{R=1}}{\phi_{R=3}}$$

$$RR_{2/3} = \frac{\phi_{R=2}}{\phi_{R=3}}$$

Relative Risk Ratios

Relative risk ratios (RRR) describe the relative risks of being in one category versus the reference category, conditional on a change in a predictor.

$$RRR_{1/3, X} = \frac{RR_{1/3} \mid X=1}{RR_{1/3} \mid X=0} = \frac{\frac{\phi_{R=1} \mid X=1}{\phi_{R=3}}}{\frac{\phi_{R=1} \mid X=0}{\phi_{R=3}}}$$

Relative Risk Ratios

You can calculate the relative risk ratio for a predictor in a multinomial logistic regression model by exponentiating its coefficient.

$$RRR_x = e^X$$

These are the odds ratios reported by HLM for coefficients in multinomial models

Relative Risk Ratio

Nice video explaining RRs and RRRs with examples and visualizations:

<https://www.youtube.com/watch?v=-cQwiEJnPyA>

IIA Assumption

This models require an assumption of **independence of irrelevant alternatives (IIA)**.

That is, the relative probability of one outcome versus another is independent of other alternatives, whether or not they exist in your data set.

IIA Assumption

For example, if predicting the relative odds of voting for a Democratic candidate versus a Republican one, we assume it doesn't matter if there is a third party candidate option.

Or if choosing to attend UChicago over Northwestern for an MA program, we assume it doesn't matter if UIC offers the same degree.

IIA Assumption

You should evaluate this assumption theoretically. Some say you can test it with a Hausman-McFadden test (see Melamed and Doan 2024 for instructions).

Other multinomial modeling techniques, such as nested logit and multinomial probit regression, allow for correlated outcomes and are more robust to the IIA assumption.

Power

Remember that your outcome variable is categorical, so your sample size is split among different categories. Multinomial logistic regression is therefore rather data-hungry.

Estimating scarce categories is particularly difficult.

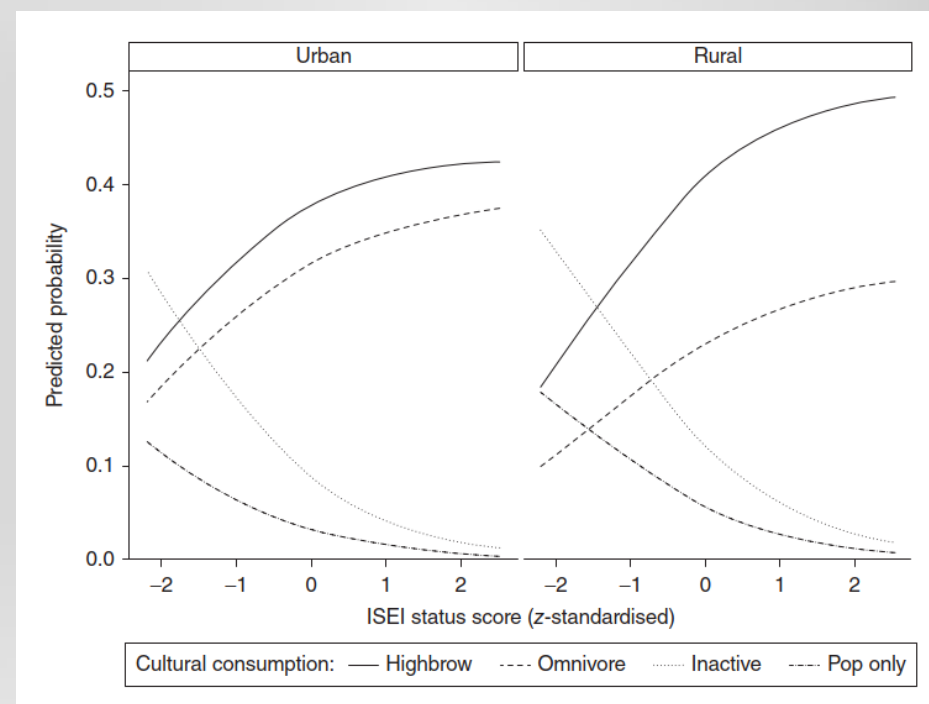
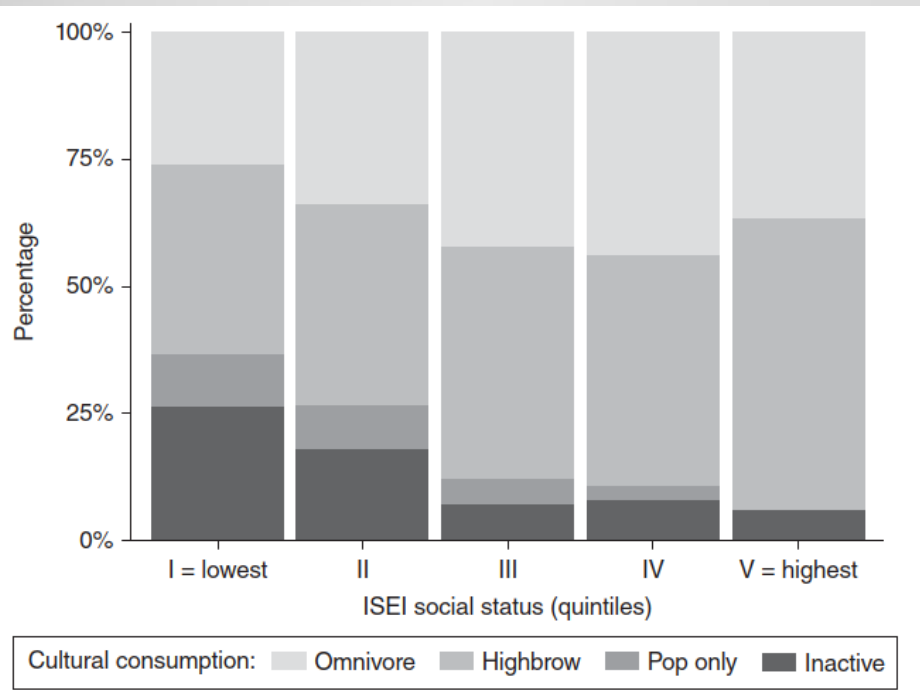
Parsimony

Because every variable you add to your model will yield multiple coefficients in your analysis, you should give extra consideration to a parsimonious model.

Give particular thought to including categorical variables, and whether they can be simplified.

Visualization

Using intelligent visualization is especially helpful in communicating effect estimates of these complex models.



Tip: Use **contrasting colors** to distinguish **qualitatively-distinct** categories, such as nominal outcomes.

Do I *Have* To?

Multinomial logit models use all information about covariates to simultaneously estimate their effects on the probability of being in different categories.

They are quite a bit more complex than ordinary logistic regression though. If you really only care about one or two categories, consider whether you can simply dichotomize your outcome.